DS-A2 — Inference and Uncertainty

Simulation-only study | Reproducible report generated from the executed notebook

Important: These results are outputs from one seeded simulation run in the current execution environment. Review the notebook and outputs yourself, perform the independent check in AI_USE_LOG.md, and write your own interpretation before submission.

1. Research question and estimand

Question: How do sample size, noise, and true mean difference affect estimation uncertainty and the probability of detecting a difference between two independent groups? The estimand is $\Delta = \mu B - \mu A$. The hypotheses are H■: $\Delta = 0$ and H■: $\Delta \neq 0$, using a two-sided $\alpha = 0.05$. The provisional minimum meaningful effect is 0.3 standard deviations.

2. Simulation and methods

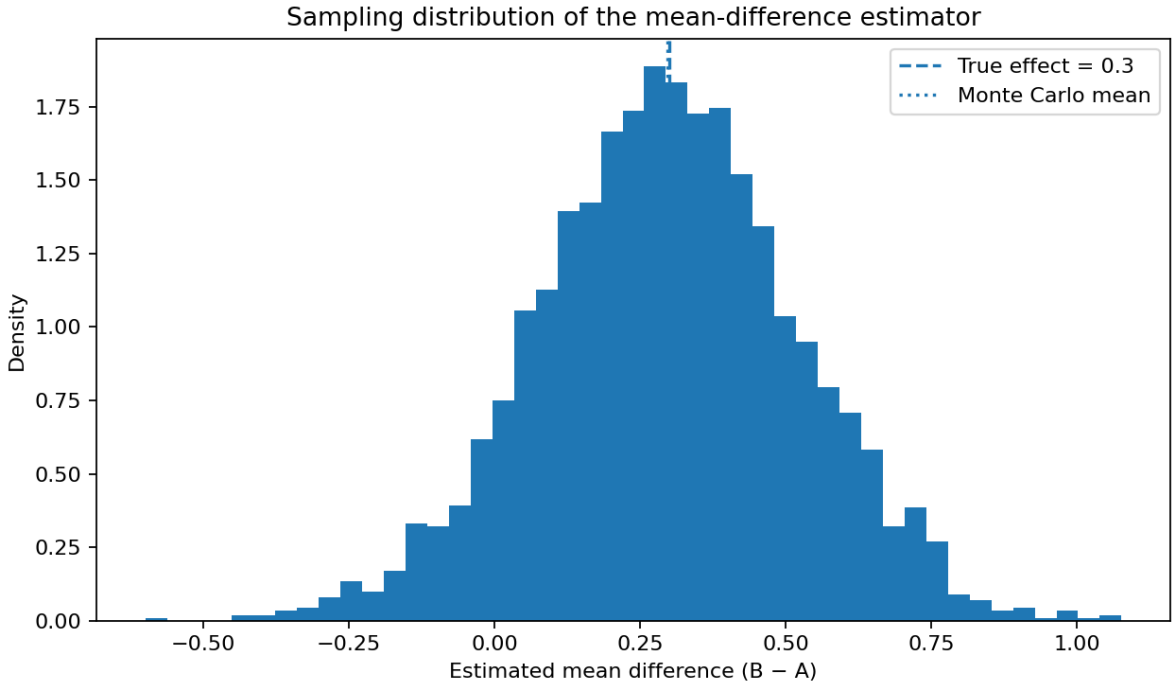
Synthetic independent samples are generated under a normal equal-variance baseline, with group A mean 0 and group B shifted by the target effect. Welch's two-sample t interval/test is the primary inferential method; a percentile bootstrap interval is a resampling comparison. The sampling distribution and power are estimated using 3,000 repetitions per setting; the bootstrap uses 4,000 resamples. The counterexample uses Student-t errors with 3 degrees of freedom, scaled to the target standard deviation.

3. Baseline estimate and intervals

method	estimate	ci_lower	ci_upper	p_value
Welch t interval/test	0.2302	-0.1864	0.6468	0.2745
Percentile bootstrap interval	0.2302	-0.1753	0.6346	nan

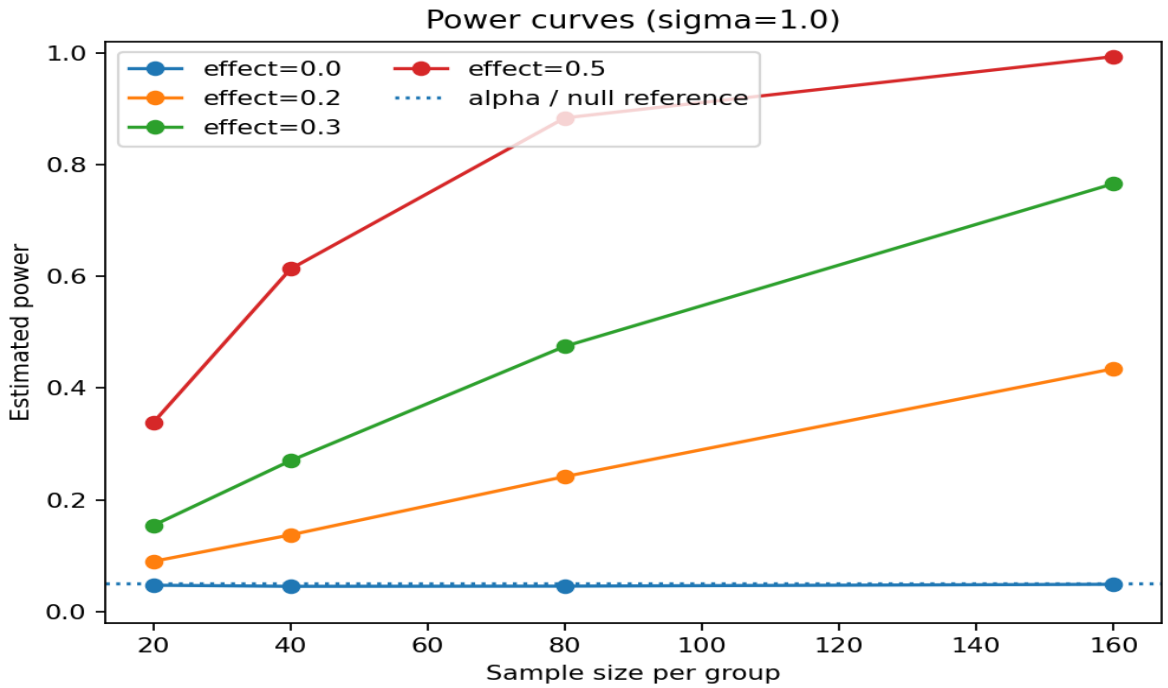
Baseline setting: n = 40 per group, true effect = 0.3, $\sigma = 1.0$. The single realization estimated a difference of 0.2302. The Welch 95% interval was [–0.1864, 0.6468] with p = 0.2745; the bootstrap interval was [–0.1753, 0.6346]. Both intervals include zero for this realization. This is one simulated sample, not evidence that the true effect is exactly zero.

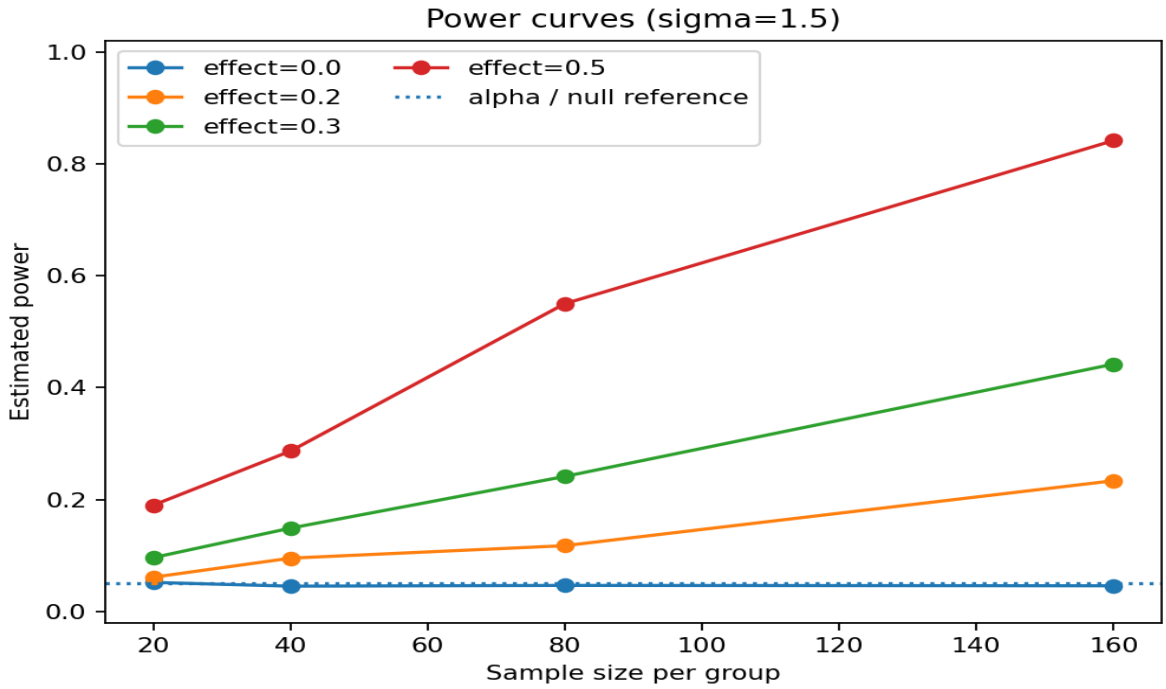
4. Sampling distribution



Across 3,000 repetitions, the Monte Carlo mean estimate was 0.2971, estimated bias was -0.0029, empirical SD was 0.2227, and theoretical SE was 0.2236. Discuss the small differences in your own words.

5. Power analysis





n_per_group	sigma	effect	rejections	estimated_power	mc_se
20.0000	1.0000	0.0000	142.0000	0.0473	0.0039
80.0000	1.0000	0.0000	137.0000	0.0457	0.0038
160.0000	1.0000	0.0000	147.0000	0.0490	0.0039
20.0000	1.0000	0.3000	463.0000	0.1543	0.0066
80.0000	1.0000	0.3000	1424.0000	0.4747	0.0091
160.0000	1.0000	0.3000	2297.0000	0.7657	0.0077
20.0000	1.5000	0.0000	156.0000	0.0520	0.0041
80.0000	1.5000	0.0000	140.0000	0.0467	0.0039
160.0000	1.5000	0.0000	138.0000	0.0460	0.0038
20.0000	1.5000	0.3000	289.0000	0.0963	0.0054
80.0000	1.5000	0.3000	724.0000	0.2413	0.0078
160.0000	1.5000	0.3000	1326.0000	0.4420	0.0091

Use the full results/power_results.csv for your comparisons. Power is conditional on the assumed model and design. At effect = 0, the rejection rate estimates the Type I error rate; departures from 0.05 should be interpreted relative to Monte Carlo uncertainty.

6. Assumption-violating counterexample

distribution	true_effect	rejection_rate	coverage_rate	mean_ci_width
normal	0.0000	0.0590	0.9410	0.8879
normal	0.3000	0.2590	0.9470	0.8874

distribution	true_effect	rejection_rate	coverage_rate	mean_ci_width
heavy_tailed	0.0000	0.0470	0.9530	0.8641
heavy_tailed	0.3000	0.3223	0.9570	0.8497

The counterexample changes the error distribution from normal to heavy-tailed Student-t while preserving finite variance. Describe what changed in the rejection rates, coverage, and interval widths. Do not generalize beyond this simulated setting.

7. Limitations and required student completion

All data are synthetic, so the results demonstrate behavior under the specified data-generating process rather than validate a real-world claim. Results are subject to Monte Carlo error; assumptions about independence, distribution shape, and group design may not hold in applications. Power depends on the assumed effect, noise, sample size, and test. Statistical significance alone does not establish practical importance.

Before submission: independently verify at least one result; complete AI_USE_LOG.md; write your own discussion of the figures and counterexample in reports/DS-A2_Report_Template.md; complete the 150–300 word platform summary; check the outputs and commit/tag the final verified version. This generated report does not replace those personal verification and reflection steps.